# Module 4 slides: Clearance and Elimination (worked examples)

Module 4 - Drug Elimination and Clearance · Lecture slides, worked examples · 14 Sep 2026

Name _____ Date _____

Five hundred mg of a drug was administered by rapid IV injection. The $V_D$ is 15 L and the elimination half-life is 8 hours. Urine samples were collected for 48 hours and 300 mg of unchanged drug was recovered.

**Slide 22** What fraction of the dose is excreted unchanged in the urine?

**Slide 22** Calculate k.

**Slides 10 and 22**

Calculate $k_e$.

**Slide 22** Calculate $Cl_T$.

## Slides 9 and 22

Calculate $Cl_R$.

## Slides 11 and 22

Calculate $Cl_H$.

**Slides 21 and 23**

A new antibiotic is actively secreted by the kidney. Its $V_D$ is 25 L in the normal adult and its clearance is 750 mL/min. What is its elimination rate constant in reciprocal hours?

**Slides 21 and 23**

A new antibiotic is actively secreted by the kidney; $V_D$ is 25 L in the normal adult. The clearance of this drug is 750 mL/min. What is the usual t½ for this drug?

**Slides 21 and 23**

The same antibiotic has a $V_D$ of 25 L. What would be the new t½ for this drug in an adult with partial renal failure whose clearance of the antibiotic was 150 mL/min?

**Slide 17** A patient is 165 cm tall. What is this height in inches?

**Slides 16 and 17**

A female patient is 165 cm tall. What is her ideal body weight?

**Slide 17** Estimate, using the ideal body weight, the creatinine clearance of a 45-year-old female who weighs 61 kg, has a serum creatinine of 1.1 mg/dL, and is 165 cm tall.